Illuminating Risk: An Ecological Analysis of Artificial Light at Night and Childhood Myopia Using State-Level US Data

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Risk Management and Healthcare Policy · 2026 · PMC13590006 · DOI: 10.2147/RMHP.S628850

Resumo (Abstract)

Purpose Childhood myopia is a growing global health concern, associated with serious vision complications later in life, including retinal detachment and myopic macular degeneration. While behavioral factors such as screen time and reduced outdoor exposure have been studied, the potential contribution of environmental light pollution, particularly artificial light at night (ALAN), remains underexplored at the population level. Methods We analyzed 2019 data from 46 US states to examine the association between ALAN and the prevalence of childhood myopia (ages 0–17). VIIRS satellite-derived nighttime radiance was used as a proxy for outdoor ambient lighting. Using generalized additive modeling (GAM), we examined nonlinear associations while adjusting for behavioral (screen time, insufficient sleep) and structural (Medicaid coverage, uninsured children, vision screening, optometrist density, urbanization) factors. Multicollinearity was assessed using Spearman correlation and variance inflation factors. Results The final GAM explained 65.3% of the deviance in childhood myopia prevalence (adjusted R 2 = 0.509). Higher ALAN was significantly associated with higher myopia prevalence (F = 8.39, p = 0.0068). Screen time showed a nonlinear relationship, whereas sleep deprivation showed an unexpected inverse association, potentially reflecting detection disparities. Medicaid coverage showed a U-shaped association, with both low and high coverage associated with higher prevalence. Vision screening rates were positively associated with myopia prevalence, likely reflecting improved detection. Urbanization, optometrist density, and the number of Asian children were not statistically significant. Conclusion ALAN is a population-level correlate of childhood myopia in the US. Although not indicative of individual-level risk, its association warrants further ecological investigation and may merit consideration in environmental health and urban planning discussions. These findings highlight the multifactorial nature of pediatric refractive error and the value of integrating environmental exposure data into public health vision surveillance. Further individual-level research is needed to assess causal mechanisms.

Introduction

Myopia, or nearsightedness, is one of the most common eye conditions globally. 1–3 Myopia results from elongation of the eyeball, causing light to focus in front of the retina and leading to blurred distance vision. 4–6 In children, myopia often progresses with age, increasing the risk of serious complications later in life, such as retinal detachment, glaucoma, and myopic macular degeneration. 7 , 8 Projections estimate that nearly half of the world’s population will be myopic by 2050, with many cases beginning in childhood. 1

While several behavioral and environmental factors, including screen time and reduced outdoor exposure, have been identified as contributors to rising myopia rates, 9–11 one factor remains underexamined in US-based ecological analyses: artificial light at night (ALAN). ALAN originates from urban streetlights, residential lighting, and commercial signage, 12 and disrupts circadian rhythms. 13 ALAN alters melatonin secretion and retinal dopamine regulation, both critical to ocular development. 14–18

ALAN exposure is rapidly increasing. 19 Between 1992 and 2017, satellite-detectable nighttime light emissions rose by at least 49%, surpassing global population growth. 20 Once considered a cosmetic or urban energy issue, ALAN is now recognized as a form of environmental pollution with potential consequences for human and ecological health. 21

Few studies have directly examined artificial light at night (ALAN) as a population-level risk factor for childhood myopia. Recent observational studies in Chinese populations have reported positive associations between higher satellite-derived outdoor ALAN exposure and both prevalent and incident childhood myopia. 22 , 23 However, this evidence remains geographically limited, and few studies have evaluated ALAN in relation to childhood myopia at the population level in other countries. In contrast, previous ecological and epidemiologic studies have more commonly focused on factors such as screen time, sleep, and healthcare access. 24–26 To address this gap, we investigated whether state-level childhood myopia prevalence was associated with ambient nighttime radiance (a satellite-derived proxy for ALAN) across the United States.

Although ALAN can be measured using satellite-derived nighttime radiance, this proxy captures ambient outdoor lighting rather than individual-level light exposure. 27 It does not reflect biologically relevant personal exposure, such as indoor lighting intensity, duration, or spectral composition, which are more directly linked to circadian regulation. 28 Still, ecological analyses using satellite data can reveal population-level patterns that generate hypotheses for future individual-level studies. 29

We apply a generalized additive modeling (GAM) framework 30 to assess this relationship while adjusting for behavioral and structural confounders. GAMs are particularly well-suited for ecological population health studies, as they allow for the flexible modeling of nonlinear associations without imposing strict parametric assumptions. 31 This method captures more nuanced exposure–outcome relationships that may be missed by traditional linear approaches, especially when multiple environmental and policy-level variables interact. 32 Our approach represents a novel application of GAM in vision epidemiology and evaluates ALAN as a potential population-level environmental correlate of childhood myopia warranting further investigation.

Material and Methods

Study Design

This ecological study examined state-level associations between environmental exposures, behavioral patterns, and structural healthcare factors with myopia prevalence in children aged 0–17 in the US The primary exposure of interest was ALAN, measured using mean nighttime radiance. Covariates were selected based on a comprehensive literature review of established myopia risk factors.

Data Sources

All data sources are listed in Table 1 . Publicly available 2019 data from 46 US states were analyzed for children aged 0–17. Alaska, Hawaii, North Dakota, and Vermont were excluded due to incomplete data from one or more sources. Table 1 Data Sources for Exposure, Outcome, and Covariate Variables Variable Data Source (Agency) Access Link Name URL ALAN(Nighttime radiance) NOAA VIIRS DNB Composites V2.1 NOAA VIIRS https://developers.google.com/earth-engine/datasets/catalog/NOAA_VIIRS_DNB_ANNUAL_V21 Myopia prevalence CDC Vision and Eye Health Surveillance System CDC VEHSS https://ddt-vehss.cdc.gov Vision screening National Survey of Children’s Health NSCH Screening

https://nschdata.org/browse/survey/allstates?q=8620 Sleep duration National Survey of Children's Health NSCH Sleep https://www.childhealthdata.org/browse/survey/archive2021/allstates?q=8362 Screen time National Survey of Children's Health NSCH Screen Time https://www.childhealthdata.org/browse/survey/archive2021/allstates?q=8325 Medicaid & Uninsured rates Kaiser Family Foundation (KFF) KFF Child Coverage https://www.kff.org/other/state-indicator/children-0-18 Asian child population US Census Bureau ACS (2019) ACS Demographics https://data.census.gov/table/ACSDP1Y2019.DP05 Optometrist density US Bureau of Labor Statistics BLS OES https://www.bls.gov/oes/2019/may/oessrcst.htm Urban population US Census Bureau (Urban/Rural Definitions) Census Geography https://www.census.gov/programs-surveys/geography/guidance/geo-areas/urban-rural.html

The principal dependent variable for our model was state-level myopia prevalence in children aged 0–17 years, collected from the Centers for Disease Control and Prevention (CDC) Vision and Eye Health Surveillance System (VEHSS). VEHSS compiles de-identified health claims from private insurance, Medicare, Medicaid, and the military health system; electronic health records (IRIS Registry); and survey data, including NHANES, ACS, and NSCH. Myopia is identified within VEHSS through the presence of the corresponding ICD-9 or ICD-10 code in health claims or electronic health records. VEHSS estimates differ from survey- and refractive-error–based measurements such as those in NHANES, which tend to yield higher prevalence due to non-cycloplegic refraction. We selected VEHSS because it provides consistent, state-level coverage, enabling ecological comparison. While this diagnostic basis may underestimate true prevalence, it ensures methodological uniformity across states.

The primary independent variable of interest was ALAN, operationalized as the annual mean nighttime radiance from the NOAA Visible Infrared Imaging Radiometer Suite (VIIRS) Day/Night Band (DNB) V2.1 composite for 2019. Data were extracted via Google Earth Engine at 15-arc-second (~500 m) resolution and aggregated by state mean radiance (nW/cm 2 /sr). Pixels flagged for cloud contamination or stray-light interference were masked using the VIIRS quality bands. As VIIRS measures outdoor ambient light, it does not capture indoor lighting or individual exposure duration, which is a limitation discussed below. 33

The National Survey of Children's Health (NSCH) provided data on three behavioral factors: 1) the percentage of children who received vision screenings, 2) the percentage of children reported to sleep less than the recommended age-appropriate hours during an average day/most weeknights, and 3) the average daily screen time, calculated as a weighted state-level average of time spent in front of a TV, computer, cellphone, or other electronic device by children. This variable was included to control for near work as a behavioral risk factor.

We used average daily screen time as a state-level proxy for near work because no publicly available dataset provides consistent measures of educational intensity or academic visual demands across states. Screen time captures a relevant behavioral domain, but it does not distinguish educational tasks from recreational digital use. Prior studies show that near-work metrics vary widely and often rely on coarse self-report instruments that lack specificity. 34 , 35 Major reviews emphasize that educational exposure represents a causal risk factor for myopia but also note the absence of standardized population-level measures that quantify reading load or academic visual demands. 11 , 36 More granular indicators of academic workload or school-based near-work duration would strengthen future ecological analyses. Exploratory sensitivity analyses incorporating state-level educational indicators, including high-school graduation rate and postsecondary enrollment, did not materially alter the association between ALAN and childhood myopia prevalence (Supplementary

Table S1).

The Kaiser Family Foundation's analysis of the American Community Survey (ACS) reported the percentages of uninsured children and of children aged 0–18 who are insured by Medicaid. The ACS also supplied the number of Asian children aged 0–19.

Optometrist counts came from the Bureau of Labor Statistics. Because 2019 data were unavailable, the 2020 US Census provided the closest estimate of urban population percentages.

Other key risk factors for myopia include parental refractive error, genetic susceptibility, and UV exposure, while time spent outdoors is considered a significant protective factor. However, no publicly available state-level data on these variables were accessible for inclusion in this analysis.

Data Preprocessing

Datasets were merged using consistent state-level identifiers. Listwise deletion was applied for states missing predictor data, except for optometrist counts, where median imputation was used to handle a small number of missing values and reduce the influence of extreme values in the right-skewed distribution. All continuous predictor variables were standardized with z-scores to facilitate interpretability. Asian child and optometrist counts were converted to percentages, but results were materially unchanged in the GAM.

Statistical Analysis

We first assessed multicollinearity using Variance Inflation Factors (VIFs); variables with VIFs>5 were excluded to reduce redundancy. Stepwise regression (bidirectional, AIC-based) identified variables that contributed meaningfully to the model fit. Initial modeling began with an ordinary least squares regression. Still, visual inspection of residuals and exploratory plots indicated potential non-linear relationships for several predictors, particularly sleep deprivation and the number of Asian children. We therefore employed a GAM to flexibly assess nonlinear associations between predictors and myopia prevalence. This approach was particularly suited for variables such as screen time and sleep deprivation, for which non-linear or threshold effects have previously been reported. 37 , 38 The final GAM was implemented using penalized regression splines, with smoothing parameters estimated via Restricted Maximum Likelihood (REML) to prevent overfitting. 39 Although a weighted model was considered to account for differences in data reliability, the unweighted model performed better in terms of AIC and R 2 . We assessed multicollinearity using Spearman correlation 40 , 41 and variance inflation factors. 42 We evaluated spatial autocorrelation in the standardized residuals using Moran's I with a first-order contiguity weights matrix.

Software

Mean radiance values were extracted from GeoTIFF files using QGIS (v3.40.1). All statistical analyses were conducted in R (v2024.09.1+394) using mgcv (v1.9.1), MASS (v7.3.61), car (v3.1.3).

Artificial Intelligence

In addition to traditional searches like PubMed, we utilized Elicit.com and Consensus.app to target specific areas for the most up-to-date evidence. We developed this manuscript using GPT-4o (OpenAI), which helped generate initial outlines, which we extensively revised to fit our planned structure. We used GPT-4o to explore structural drafts and subheading phrasing, which we subsequently revised and grounded in empirical data. We rewrote these drafts to articulate our ideas

fully and include the necessary citations. We used GPT-4 and GPT-4o throughout the writing process to periodically review and enhance grammar and sentence structure. In alignment with recommended guidelines for the ethical use of AI in research and publishing, 43 we affirm full accountability for the accuracy, originality, and scientific validity of the content in this manuscript. While AI tools supported evidence synthesis, we independently conducted all key evaluations, interpretations, and revisions to ensure the work adheres to current scientific knowledge and meets rigorous academic standards.

Results

Multivariable Modeling Reveals Population-Level Predictors of Pediatric Myopia

This study evaluated whether artificial light at night (ALAN) may act as a population-level risk factor for childhood myopia in the United States. To assess this relationship, we employed a generalized additive model (GAM), which allowed us to examine potential nonlinear associations while adjusting for a range of covariates. These included environmental exposures (eg, screen time, sleep deprivation), behavioral patterns, and healthcare access indicators such as vision screening prevalence and Medicaid coverage rates.

The purpose of the model was not to predict individual-level myopia prevalence, but to test whether the hypothesized ecological relationship between ALAN and myopia persisted after accounting for these potential confounders. This analytical approach is particularly well-suited for population-level analyses where exposures may not have a linear relationship with outcomes.

In unadjusted analyses including only ALAN, the association with childhood myopia remained significant ($F = 4.64$, $p = 0.0386$). In the fully adjusted GAM, the effect persisted ($F = 8.39$, $p = 0.0068$), demonstrating robustness to confounder inclusion, as well as explaining 65.3% of the deviance in reported childhood myopia prevalence across 46 US states, with an adjusted R^2 of 0.509. This suggests that the included predictors collectively account for a substantial proportion of the variation in myopia rates at the state level, supporting the hypothesis that broader environmental and systemic factors may influence refractive development in children. The final generalized additive model results are summarized in Table 2 . Table 2 Summary of Generalized Additive Model Results Predictor edf F-statistic p-value Association Pattern Mean nighttime radiance (nW/cm 2 /sr) 1.000 8.39 0.0068 Linear (+) Vision screening prevalence (%) 1.000 5.16 0.0299 Linear (+) Insufficient sleep (%) 1.000 10.54 0.0027 Inverse (-) Screen time (h/day) 2.992 2.90 0.0361 Nonlinear (non-monotonic, ∩) Uninsured children (%) 1.000 0.93 0.3426 NS* Medicaid coverage (%) 2.635 3.59 0.0219 U-shaped Asian children count 1.000 1.09 0.3054 NS* Optometrist count 1.496 0.27 0.6953 NS* Urbanization (%) 1.000 0.77 0.3873 NS* Notes : + = positive/increasing association; - = negative/decreasing association; ∩ = inverted U-shaped (non-monotonic) association; NS* = Not statistically significant ($p > 0.05$).

Nighttime Light Exposure is Independently Associated with Higher Pediatric Myopia Rates

Mean nighttime radiance, used here as a proxy for ambient artificial light at night (ALAN), demonstrated a statistically significant positive association with childhood myopia prevalence across states (Figure 1A ; edf = 1.00, $F = 8.39$, $p = 0.0068$). This indicates a linear relationship between increasing nighttime light exposure and higher reported rates of pediatric myopia, without accounting for risk factors like parental refractive status, genetic predisposition, UV exposure, and protective

factors like time spent outdoors. Figure 1 Partial effect plots from a generalized additive model (GAM) assessing state-level predictors of childhood myopia prevalence in the United States. Each panel displays the smoothed, adjusted association between a single predictor and childhood myopia prevalence across 46 US states, derived from a multivariable GAM that included environmental, behavioral, and healthcare access-related covariates. Solid black lines represent the estimated partial effect of each predictor on myopia prevalence while holding other variables constant. Shaded gray bands indicate 95% confidence intervals, reflecting uncertainty around each smoothed function. (A) shows a positive linear relationship between mean nighttime radiance (a proxy for artificial light at night) and myopia prevalence. (B) reveals a nonlinear association between average daily screen time and myopia, with elevated prevalence at moderate screen use levels and a plateau at higher exposures. (C) shows no substantial association between urban population percentage and myopia rates. (D) demonstrates a U-shaped relationship between Medicaid coverage and myopia, with the lowest prevalence at moderate coverage levels and higher prevalence at both extremes. (E and F) display inverse or unexpected associations: higher rates of uninsured children and insufficient sleep were associated with lower reported myopia prevalence, potentially reflecting detection disparities or residual confounding. (G) shows a positive association between vision screening prevalence and reported myopia, consistent with improved case detection. (H and I) (Asian population and optometrist density, respectively) did not show statistically significant associations within the model. All predictors are modeled on their original scale and reflect state-level ecological relationships. These plots illustrate the independent contribution of each variable to myopia prevalence after adjusting for all others in the model. Nine single-line plots assess childhood myopia prevalence against state-level predictors. Each plot features a black trend line with a gray confidence band and rug marks indicating data density. Image A: Mean Radiance (nW/cm superscript 2/sr) shows a positive linear trend. Image B: Screen Time (hours/day) peaks around 1 hour then declines. Image C: Urbanization (%) shows a flat trend. Image D: Medicaid Coverage (%) displays a U-shaped curve. Image E: Uninsured Children (%) shows a slight negative trend. Image F: Insufficient Sleep (%) declines steadily. Image G: Vision Testing (%) shows a positive trend. Image H: Asian Children Count (%) shows a positive trend. Image I: Optometrist Count (%) has a slight hump. Overall, predictors like mean radiance and vision testing show positive associations, while insufficient sleep shows a negative association. Medicaid coverage has a U-shaped relationship and urbanization shows no substantial association. Nine single-line plots showing childhood myopia prevalence against state-level predictors in the United States.

Our results show that states with greater levels of ALAN tended to report elevated childhood myopia rates. This relationship persisted after adjusting for behavioral factors and access to care, suggesting that ALAN may contribute to population-level disparities in refractive development.

The association is biologically plausible, given existing evidence that ALAN can suppress melatonin production and disrupt circadian signaling, both of which are implicated in ocular growth regulation. Children exposed to excessive nighttime light may experience prolonged retinal stimulation and altered dopaminergic activity, which can influence axial length and the progression of refractive errors.

Importantly, ALAN represents a potentially modifiable environmental exposure warranting further investigation regarding pediatric eye health. The observed association highlights the relevance of environmental exposures in shaping pediatric eye health outcomes.

Behavioral Predictors Show Paradoxical Sleep Association and Nonlinear Screen Time Effects

The proportion of children experiencing insufficient sleep demonstrated a statistically significant negative association with childhood myopia prevalence (Figure 1F ; edf = 1.00, F = 10.54, p = 0.0027). Surprisingly, states where more children reported inadequate sleep tended to have lower reported rates of myopia. This counterintuitive association is likely an ecological artifact reflecting disparities in diagnostic access and healthcare infrastructure, rather than a biologically protective effect (Figure 1F). Caution is warranted in interpreting this result. States with higher rates of sleep deprivation may also have limited pediatric vision screening, fewer school-based eye exams, or lower parental awareness, leading to underdiagnosis of refractive error among more vulnerable populations.

This pattern may also indicate residual confounding by socioeconomic status or educational intensity, as higher-income states, where children may sleep longer due to more stable home environments, tend to have better screening infrastructure and earlier detection of myopia. While some individual-level studies suggest poor sleep may accelerate myopic progression, our ecological findings emphasize the complexity of measuring sleep as a population-level risk factor and underscore the importance of context in interpreting behavioral data.

In contrast, average screen time among children exhibited a nonlinear association with myopia prevalence (Figure 1B ; edf = 2.99, F = 2.90, p = 0.0361). States with moderate levels of screen exposure showed elevated myopia rates, while those with very high usage did not show further increases. This plateau effect (Figure 1B) may reflect behavioral compensation (eg, increased outdoor time in high-use households), measurement limitations, or detection ceiling effects. While screen time is widely discussed as a myopia risk factor, our findings suggest its impact may be modulated by other state-level factors such as lighting conditions, education intensity, and time outdoors, which were not directly measured in this study.

Insurance Coverage and Screening Intensity Shape Reported Myopia Patterns

Medicaid coverage exhibited a statistically significant U-shaped relationship with childhood myopia prevalence (Figure 1D ; edf = 2.64, F = 3.59, p = 0.0219). States with moderate coverage levels (~40–50%) had the lowest predicted prevalence, while both lower (<30%) and higher (>60%) coverage levels were associated with increased rates (Figure 1D). This trend persisted after controlling for sleep (Figure 1F), screen time, and other demographic factors. This U-shaped association suggests that the relationship between insurance coverage and reported myopia is complex and may not follow a simple linear gradient. Although the GAM model does not establish causality, the pattern suggests that access and the effectiveness of translating coverage into care delivery likely influence detection and diagnosis rates at the state level. Vision screening prevalence also demonstrated a statistically significant positive linear association with myopia rates (Figure 1G ; edf = 1.00, F = 5.16, p = 0.0299). States conducting more frequent or comprehensive vision screenings reported a higher prevalence of childhood myopia. This likely reflects detection differences (Figure 1G), not increased disease incidence. States with stronger school health mandates and infrastructure may identify more cases, particularly among younger or previously unexamined populations.

These findings highlight the central role of insurance coverage and surveillance intensity in shaping reported health outcomes and suggest that myopia prevalence, like many pediatric conditions, is partly a function of the visibility and accessibility of care.

Non-Significant Predictors Still Contribute to Model Robustness

Several covariates included in the model to control for structural and demographic variability did not demonstrate statistically significant associations with childhood myopia prevalence. These included the percentage of uninsured children (Figure 1E ; $p = 0.343$), optometrist density per 100,000 population (Figure 1I ; $p = 0.695$), number of children of Asian descent (Figure 1H ; $p = 0.305$), and the proportion of the population residing in urban areas (Figure 1C ; $p = 0.387$).

Although these factors were not individually associated with state-level myopia prevalence within the GAM framework, their inclusion was critical. They helped control for healthcare access variability, population density effects, and ethnically patterned risk distributions that may confound associations between environmental or policy-level exposures and pediatric vision outcomes.

For instance, while optometrist density was insignificant, it remains a relevant contextual factor influencing diagnosis capacity. Similarly, while the percentage of children of Asian descent is known to correlate with myopia risk at the individual level, its state-level influence may be diluted due to aggregation or co-variation with other variables such as Medicaid coverage and educational attainment.

Inclusion of these covariates contributes to model robustness by reducing residual confounding and ensuring that associations identified for ALAN, sleep, screen time, and Medicaid reflect net effects after adjusting for broader demographic and healthcare infrastructure differences. Table 3 summarizes the descriptive statistics of these variables. Table 3 Descriptive Statistics of Exposure, Outcome, and Covariate Variables Variable Mean SD Min Max Source Myopia prevalence (%) 35.43 5.87 20.60 49.30 2019 CDC VEHSS Mean nighttime radiance (nW/cm 2 /sr) 1.90 1.57 0.33 7.27 2019 NOAA VIIRS Vision screening prevalence (%) 62.6 3.94 54.70 72.50 2019 NSCH Insufficient sleep (%) 35.1 5.57 26.60 50.50 2019 NSCH Screen time (h/day) 2.17 0.09 2.01 2.37 2019 NSCH Uninsured children (%) 5.23 2.22 1.30 12.80 2019 ACS Medicaid coverage (%) 36.6 7.63 15.30 55.60 2019 ACS Asian children count 108365 222,144 2853 14,456,226 2019 ACS Optometrist count 55.78 14.83 24.00 101 2019 BLS Urbanization (%) 73.36 14.10 38.63 94.24 2020 US Census Abbreviations : AASM, American Academy of Sleep Medicine; ACS, American Community Survey; ACF, autocorrelation function; AI, artificial intelligence; AIC, Akaike information criterion; ALAN, artificial light at night; BLS, Bureau of Labor Statistics; CDC, Centers for Disease Control and Prevention; DNB, Day/Night Band; GAM, generalized additive model; ICD, International Classification of Diseases; IRIS, Intelligent Research in Sight Registry; KFF, Kaiser Family Foundation; NHANES, National Health and Nutrition Examination Survey; NOAA, National Oceanic and Atmospheric Administration; NS, not statistically significant; NSCH, National Survey of Children’s Health; QGIS, Quantum Geographic Information System; REML, restricted maximum likelihood; SD, standard deviation; SES, socioeconomic status; VEHSS, Vision and Eye Health Surveillance System; VIF, variance inflation factor; VIIRS, Visible Infrared Imaging Radiometer Suite.

Model Diagnostics Confirm Assumptions of Validity and Independence

Diagnostic plots, presented in Figure 2 , confirmed that the generalized additive model (GAM) met the key assumptions necessary for valid inference. The residuals vs fitted values plot (Figure 2A)

displayed no systematic pattern or heteroscedastic spread, suggesting that the variance of residuals remained relatively constant across the range of fitted values. This supports the assumption of homoscedasticity. The quantile–quantile (Q-Q) plot (Figure 2B) indicated that standardized residuals closely followed the expected normal distribution, with only mild deviation at the extreme tails. These small departures do not materially affect the interpretability of model coefficients, particularly in ecological studies where minor non-normality is common. The histogram of residuals (Figure 2C) showed an approximately symmetric distribution with a slight right skew. While not perfectly normal, the shape remains within acceptable boundaries for GAM interpretation and is unlikely to bias results given the model's additive structure. The autocorrelation function (ACF) plot (Figure 2D) revealed no significant autocorrelation among residuals at any lag, confirming that the residuals were independent across observations. Spatial autocorrelation testing of standardized residuals using Moran's I (based on state centroids) yielded no significant spatial dependence ($I = 0.041$, $p = 0.2459$), indicating that residuals were geographically independent (Supplementary Figure S1). This is especially important in geographically structured data where spatial dependence could otherwise bias inferences. Collectively, these diagnostic assessments support the model's validity and indicate that the reported associations between predictors and childhood myopia prevalence are not likely to be artifacts of assumption violations. Figure 2 Model diagnostic plots evaluating assumptions of the generalized additive model (GAM) used to assess state-level predictors of childhood myopia prevalence. These four panels evaluate the validity of key assumptions underlying the fitted GAM: (A) (Residuals vs Fitted): Residuals appear randomly scattered around zero, with no clear pattern or funnel shape, suggesting homoscedasticity (constant variance) across the range of fitted values. (B) (Q-Q Plot of Residuals): Standardized residuals closely follow the expected normal distribution, with only minor deviation at the extreme tails. This supports the assumption of approximate residual normality. (C) (Histogram of Residuals): The distribution of residuals is approximately symmetric but shows mild right skew. The shape remains within acceptable limits for linear approximation, especially in an ecological, population-level analysis. (D) (Autocorrelation Function – ACF): No significant autocorrelation is detected among residuals. The lags fall within 95% confidence bounds, confirming Independence of observations across states. Together, these diagnostics suggest that the fitted GAM meets the key modeling assumptions required for valid inference and supports the robustness of the reported associations. Image A: Scatter plot of Residuals vs Fitted Values. X-axis: Fitted Values (25-45), Y-axis: Residuals (-5 to 10). Reference line at Residuals = 0. Points range from Fitted Values 23-45 and Residuals -8 to 10. Image B: Quantile plot with a diagonal reference line. X-axis: x (-2 to 2), Y-axis: y (-5 to 10). Points span from (-2, -8) to (2, 10). Image C: Histogram of Residuals. X-axis: Residuals (-5 to 10), Y-axis: Frequency (0.0 to 10.0). Bars cluster around Residuals 0 to 3, with smaller bars from -7 to 11. Image D: Autocorrelation function plot. X-axis: Lag (5 to 20), Y-axis: ACF (-0.3 to 0.3). Vertical lines at integer lags, with dashed lines near 0.3 and -0.3. Largest positive spike at lag 12 (~0.22), largest negative spike at lag 6 (~-0.26). Four plots showing residuals versus fitted, a quantile plot, a residuals histogram and autocorrelation.

Low Predictor Collinearity Supports Model Integrity and Interpretability

We conducted a pairwise Spearman correlation analysis to assess potential multicollinearity among predictors, as visualized in Figure 3 . The matrix illustrates the direction and strength of association between all included covariates. The strongest observed correlation was between Medicaid coverage and insufficient sleep ($\rho = 0.51$), indicating a moderate positive association. This relationship may reflect broader underlying socioeconomic or structural dynamics: for example, states with higher Medicaid enrollment may also have higher rates of sleep disruption in children due to housing

instability, caregiver shift work, or other contextual stressors. All other pairwise correlations remained below |0.50|, most falling under |0.40|. These low to moderate correlations suggest that the included predictors are relatively independent at the state level and unlikely to introduce significant multicollinearity into the GAM model. These findings support the results of our preliminary variance inflation factor (VIF) analysis, which found no predictors exceeding standard thresholds for concern about collinearity. Together, these diagnostics increase confidence that the reported associations reflect independent contributions of each variable rather than redundant or structurally linked effects. Figure 3 Spearman correlation matrix for the nine standardized predictors included in the generalized additive model (GAM). Each cell displays the Spearman correlation coefficient (ρ) between a pair of predictors, quantifying the direction and strength of monotonic relationships. Color shading represents the magnitude and sign of each coefficient, with blue indicating positive associations and red indicating negative associations. Stronger shading corresponds to higher absolute correlation values. The matrix confirms that most predictor pairs have low to moderate correlation, with only Medicaid coverage and insufficient sleep exceeding a coefficient of |0.50| (ρ = 0.51). All other correlations fall below standard multicollinearity thresholds, supporting the assumption that the included variables contribute independently to the GAM. These findings align with the preliminary variance inflation factor (VIF) analysis results. This visual diagnostic supports the model's integrity and demonstrates minimal structural overlap between predictors, ensuring valid interpretation of partial effects in the multivariable framework. A triangular Spearman correlation heatmap with numeric coefficients printed in each shown cell. Row labels: Asian Children Count percent■ Vision Testing percent■ Insufficient Sleep percent■ Uninsured Children percent■ Medicaid Coverage percent■ Urbanization percent■ Screen Time h slash day■ Mean Radiance nW slash cm superscript 2 slash sr. Column labels: Screen Time h slash day■ Urbanization percent■ Medicaid Coverage percent■ Uninsured Children percent■ Insufficient Sleep percent■ Vision Testing percent■ Asian Children Count percent■ Optometrist Count. A legend labeled, Correlation, ranges from minus 1.0 through 0.0 to 1.0■ the legend indicates negative correlations at the minus end, positive correlations at the plus end and stronger shading for larger absolute values. Most coefficients are near zero in magnitude. The largest positive coefficient shown is Medicaid Coverage percent with Insufficient Sleep percent equals 0.51. Other notable positive values include Mean Radiance nW slash cm superscript 2 slash sr with Urbanization percent equals 0.36■ Medicaid Coverage percent with Vision Testing percent equals 0.32■ Mean Radiance nW slash cm superscript 2 slash sr with Insufficient Sleep percent equals 0.27■ Insufficient Sleep percent with Vision Testing percent equals 0.26■ Mean Radiance nW slash cm superscript 2 slash sr with Vision Testing percent equals 0.26■ Screen Time h slash day with Medicaid Coverage percent equals 0.21■ Urbanization percent with Optometrist Count equals 0.17■ Screen Time h slash day with Urbanization percent equals 0.16■ Screen Time h slash day with Uninsured Children percent equals 0.16■ Vision Testing percent with Optometrist Count equals 0.16■ Vision Testing percent with Asian Children Count percent equals 0.15■ Uninsured Children percent with Insufficient Sleep percent equals 0.15. Notable negative values include Urbanization percent with Insufficient Sleep percent equals minus 0.28■ Urbanization percent with Medicaid Coverage percent equals minus 0.24■ Mean Radiance nW slash cm superscript 2 slash sr with Uninsured Children percent equals minus 0.18■ Urbanization percent with Uninsured Children percent equals minus 0.16■ Insufficient Sleep percent with Asian Children Count percent equals minus 0.15■ Uninsured Children percent with Vision Testing percent equals minus 0.14■ Asian Children Count percent with Optometrist Count equals minus 0.14■ Mean Radiance nW slash cm superscript 2 slash sr with Asian Children Count percent equals minus 0.12. Remaining shown coefficients: Screen Time h slash day with Vision Testing percent equals minus 0.08; Medicaid Coverage percent with Optometrist Count equals minus 0.07; Medicaid Coverage percent with

Uninsured Children percent equals minus 0.06■ Mean Radiance nW slash cm superscript 2 slash sr with Screen Time h slash day equals minus 0.05■ Screen Time h slash day with Insufficient Sleep percent equals minus 0.03■ Uninsured Children percent with Optometrist Count equals minus 0.03■ Screen Time h slash day with Optometrist Count equals 0■ Uninsured Children percent with Asian Children Count percent equals 0.01■ Urbanization percent with Asian Children Count percent equals 0.03■ Screen Time h slash day with Asian Children Count percent equals 0.04■ Mean Radiance nW slash cm superscript 2 slash sr with Optometrist Count equals 0.04■ Mean Radiance nW slash cm superscript 2 slash sr with Medicaid Coverage percent equals 0.08; Medicaid Coverage percent with Asian Children Count percent equals 0.11■ Insufficient Sleep percent with Optometrist Count equals 0.12■ Urbanization percent with Vision Testing percent equals 0.12. A Spearman correlation heatmap with labeled rows and columns and numeric coefficients in cells.

Figure 4 visually synthesizes our final model, distilling nonlinear patterns and domain-specific contributions into a single framework. Unlike the statistical tables, the figure spatially arranges predictors to underscore how environmental, behavioral, and structural factors jointly shape the prevalence of pediatric myopia. ALAN and vision screening emerge as positive linear contributors, while Medicaid coverage and screen time follow complex curvilinear patterns. By mapping these associations around a symbolic ocular structure, the figure reinforces the biological relevance of our outcome and clarifies where population-level interventions might reduce risk. Figure 4 Conceptual summary of ecological predictors of childhood myopia prevalence across 46 US states. This figure illustrates the multivariable associations identified in a generalized additive model (GAM) evaluating state-level predictors of childhood myopia prevalence. Arrows represent the direction and shape of each relationship: a positive linear association for artificial light at night (ALAN) and vision testing; a U-shaped association for Medicaid coverage; a nonlinear, non-monotonic association (∩) for screen time; and an inverse (negative linear) association for sleep deprivation. Covariates, including urban setting, ethnic profile, and eye care access, were not statistically significant but were retained in the model to adjust for structural and demographic confounders. Color-coded domains represent environmental (green), behavioral (red), and healthcare/structural (blue) factors, as well as non-significant effects (gray). The central panel depicts childhood myopia as a graphical eye structure with a retinal overlay, emphasizing the anatomical and developmental relevance of the outcome. This synthesis highlights the ecological complexity of pediatric myopia and the importance of modifiable population-level exposures. Created in BioRender (https://BioRender.com). A diagram illustrating factors influencing childhood myopia prevalence. Environmental factors include ALAN (light pollution) with a linear association. Healthcare and structural factors show Medicaid coverage with a U-shaped association and vision screening with a linear association. Behavioral factors include screen time with a nonlinear association and sleep deprivation with an inverse association. Covariates such as urban setting, ethnic profile and optometrist density are noted as non-significant. The central panel depicts an eye with the text, ′Eyeball is too long. Light is focused in front of the retina.′ Arrows indicate the direction and type of associations: linear positive, inverse negative, nonlinear and U-shaped. Diagram of factors affecting childhood myopia prevalence with associations.

Discussion

Our ecological analysis identified artificial light at night (ALAN), sleep deprivation, screen time, Medicaid vision coverage, and vision screening access as significant predictors of childhood myopia prevalence at the state level. In contrast, uninsured rates, optometrist count, Asian ethnicity, and urbanization were not statistically significant in this study.

Artificial Light at Night (ALAN)

ALAN exposure emerged as one of the strongest predictors in our model, showing a positive linear relationship with myopia prevalence without incorporating contributors like familial refractive history, genetic risk, UV light exposure, and mitigating factors such as time spent outdoors. While earlier US-based studies failed to find significant associations between nighttime lighting and myopia, 44 , 45 more recent international work has observed dose-dependent or nonlinear effects of ALAN on refractive development. 23 Our study extends prior work by evaluating satellite-derived ambient nighttime radiance in relation to childhood myopia at the population level in the United States and by examining this association alongside behavioral, demographic, and healthcare-related factors within a generalized additive modeling framework.

Mechanistically, chronic exposure to ALAN may affect ocular growth through circadian disruption. Ambient light conditions modulate dopamine and melatonin, two light-sensitive neuromodulators that regulate axial elongation. 14 , 46 Disruption of natural light-dark cycles may blunt their protective effects during critical periods of visual development.

Although our cross-sectional design limits causal inference, the observed associations support further exploration of ALAN in myopia risk models. These findings should be interpreted as hypothesis-generating. Future studies should explore individual-level light exposure using wearable devices, assess the impact of light spectra and intensity, and evaluate temporal exposure patterns. Additionally, ALAN may reflect broader characteristics of urban environments that were not fully captured in our model. Residual socioeconomic confounding and co-occurring environmental exposures may also have contributed to the observed association between ALAN and childhood myopia. For example, ALAN may correlate with broader features of the urban environment, including air pollution, greenspace availability, housing density, and other socioeconomic characteristics not fully captured in our ecological model. Accordingly, ALAN's independent contribution should be interpreted with caution. Future individual-level and multilevel studies incorporating more granular socioeconomic and environmental exposure data will be important for disentangling these interrelated factors.

Sleep Deprivation

Sleep deprivation was the strongest predictor in our model, although its inverse relationship with myopia prevalence was unexpected. This finding contrasts with longitudinal and experimental literature, which has consistently identified poor sleep hygiene as a risk factor for myopia onset and progression. 38 Some studies have reported null or inconsistent associations, potentially due to population heterogeneity or variability in sleep measurement. 26 , 47–49

As with ALAN, circadian regulation of retinal dopamine and melatonin provides a biologically plausible pathway, although our study cannot establish timing or directionality. 50–53 However, in our ecological context, this inverse association is likely an artifact of confounding and underdiagnosis rather than a true protective effect. States with higher proportions of sleep-deprived children may also reflect broader socioeconomic challenges, including increased poverty rates and reduced access to pediatric vision care, both of which can contribute to underdiagnosis. 54 , 55 As with other predictors, individual-level inference is limited due to the study design.

This result should not contradict the existing evidence linking insufficient sleep to myopia risk at the individual level. Instead, it highlights how structural disparities may obscure actual disease burden in population-level analyses. Future longitudinal studies using objective sleep and diagnostic measures

are needed to clarify the nature of this relationship.

Screen Time

Screen time showed a nonlinear association with childhood myopia prevalence. Moderate use was associated with increased myopia rates, but this risk plateaued at higher durations. This may reflect a saturation point for digital eye strain or behavioral adaptation, such as switching to passive activities with less near-focus demand. 56

Measurement limitations likely contributed to this plateau effect. For example, self-reported screen time may conflate passive exposure with more intensive near work, obscuring dose-response relationships. Prioritizing device-specific logging and distinguishing passive from active screen use will help clarify whether an actual exposure threshold exists. Nevertheless, this finding adds to an extensive literature connecting excessive screen use to childhood myopia. It reinforces calls for public health interventions promoting healthy screen habits, including breaks, appropriate lighting, and screen-free alternatives.

Medicaid Vision Coverage

Medicaid vision coverage displayed a U-shaped association with myopia prevalence. Private insurance may dominate in low-coverage states, potentially enabling timely diagnoses for children with adequate resources. 24 , 57 , 58 In moderate-coverage states, mixed-payer systems may result in fragmented care and inconsistent screening, potentially delaying diagnosis among underinsured populations. Among these interpretations, delayed diagnosis in fragmented care settings may be the most plausible, though further research is needed to test this pathway. In high-coverage, low-SES states, greater recorded prevalence may reflect improved access to care and higher rates of advanced disease prompting care-seeking. 58

This U-shaped pattern should be interpreted as exploratory. It may reflect a complex interplay between insurance coverage, service delivery capacity, and population risk distribution, but further study is needed to confirm these dynamics. Given the limitations of aggregate data and our cross-sectional design, these interpretations should be viewed with caution. Because Medicaid enrollment often serves as a proxy for lower-income status, this finding underscores the importance of equitable access to pediatric vision care across the socioeconomic spectrum, not only through coverage itself, but also through provider availability, screening programs, and care integration.

Vision Screening Prevalence

Vision screening prevalence was positively associated with myopia prevalence, likely reflecting improved case detection. States with more robust screening infrastructure may identify cases earlier or more consistently. 59

However, significant variability exists in how vision screening is mandated and implemented across states. National guidelines on frequency, referral, and follow-up care could help standardize practices and reduce geographic disparities in detection and treatment. Without such standardization, screening disparities may widen existing inequities in pediatric vision care, especially in lower-resourced states.

Non-Significant Predictors

Urbanization, the number of Asian children, optometrist density, and uninsured rates were not significant predictors. This may reflect ecological limitations, such as state-level aggregation masking within-state clustering. For example, the null result for Asian ethnicity, despite its well-established individual-level association with myopia, likely reflects data dilution rather than a true absence of effect. 60 , 61

The geographic variation in ALAN and its association with myopia risk may also reflect broader environmental justice concerns. 62 Communities with higher exposure to light pollution often overlap with socioeconomically disadvantaged areas, compounding barriers to vision health. 62 These patterns underscore the need for equity-focused urban planning and vision surveillance strategies, particularly in regions where environmental exposures and access disparities intersect. Although we modeled predictors independently, these behavioral and environmental exposures likely interact. High ALAN environments may also experience shorter sleep durations and potentially more screen exposure, although the latter relationship requires further investigation. 63–65 Future studies should explore these overlapping risks using multilevel or longitudinal designs to understand these complex dynamics.

Strengths and Limitations

To our knowledge, this is the first US-based ecological study to apply generalized additive modeling to childhood myopia, with a primary focus on artificial light at night (ALAN) as an environmental exposure. By explicitly modeling ALAN alongside behavioral and healthcare access variables, our study provides novel insight into an underexplored risk factor. The model demonstrated strong explanatory power (adjusted $R^2 = 0.509$; deviance explained = 65.3%), supporting the utility of state-level predictors in understanding population-level variation.

Nonetheless, several limitations should be acknowledged. The cross-sectional, ecological design limits causal inference and raises the possibility of ecological fallacy. Many predictor variables, including screen time, sleep, and vision screening prevalence, were based on self- or parent-reported survey data, which are subject to recall bias and reporting inaccuracies. Our outcome measure of childhood myopia prevalence was based on aggregated CDC VEHSS estimates for children aged 0–17, without stratification by narrower age bands. Since myopia onset and progression vary significantly by developmental stage, age aggregation may obscure important subgroup differences.

Additionally, because VIIRS quantifies outdoor radiance, it omits indoor illumination and personal exposure patterns that more directly affect circadian physiology. This ecological proxy is thus best interpreted as an indicator of environmental context rather than direct biological dose. Furthermore, it is important to note that ecological data cannot define a safe exposure threshold.

Slight inconsistencies in age definitions across data sources (eg, 0–17 vs 0–19 years) may also introduce minor measurement error. The sleep deprivation variable was derived using AASM-based thresholds but applied as a statewide estimate spanning infancy through adolescence, limiting developmental specificity.

Additionally, several key individual-level risk factors, such as parental refractive status, genetic predisposition, and ultraviolet light exposure, as well as protective factors such as time spent outdoors, could not be included due to limitations of publicly available data. 57 , 66–69 The absence of these variables limits the model’s completeness and the strength of causal interpretations. Future research should aim to incorporate these individual-level variables through more granular datasets.

Moreover, lighting infrastructure, housing density, and pediatric care systems differ internationally; thus, the generalizability of US results should be evaluated through replication in other regions using harmonized exposure metrics.

Despite these limitations, our analysis provides a strong foundation for future research leveraging more granular, individual-level data and highlights the potential relevance of ALAN in shaping pediatric eye health.

Future Directions

ALAN exposure may be related to myopia, and further research into its specific features could inform future policy. Low-emission street lighting and access to green spaces have been hypothesized as potentially protective. 58 Identifying which aspects of ALAN, such as intensity, spectrum, or timing, are most relevant to myopia risk may help guide city planning and zoning decisions. However, individual-level studies using objective light exposure data are still needed to justify structural changes and establish clearer causal pathways.

Conclusion

This ecological analysis identifies artificial light at night (ALAN) as a significant and underexamined environmental correlate of childhood myopia in the United States. Using satellite-derived VIIRS radiance data, we found that higher nighttime light exposure was associated with increased state-level childhood myopia prevalence. 19 , 21 , 23 While earlier US studies examined indoor nighttime lighting during early childhood with mixed results, 42 , 43 our approach offers a broader population-level perspective on community-wide ambient exposure.

Although our cross-sectional design precludes causal inference, several additional socioeconomic and behavioral factors, including screen time, Medicaid vision coverage, vision screening rates, and sleep deprivation, were also associated with state-level childhood myopia prevalence, highlighting the complex and multifactorial nature of childhood myopia.

Further individual-level, longitudinal studies are needed to clarify causal mechanisms and validate the ecological associations observed in this study. Future longitudinal studies should incorporate individual-level measures of ALAN exposure, including intensity, spectral composition, timing, and duration, to clarify its biological relevance and potential role in the development of childhood myopia. 14 , 46